INCH-POUND

MIL-DTL-17/131C
19 June 2015
SUPERSEDING
MIL-C-17/131B
31 March 1992

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, TRIAXIAL, 50 OHMS, M17/131-RG403

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

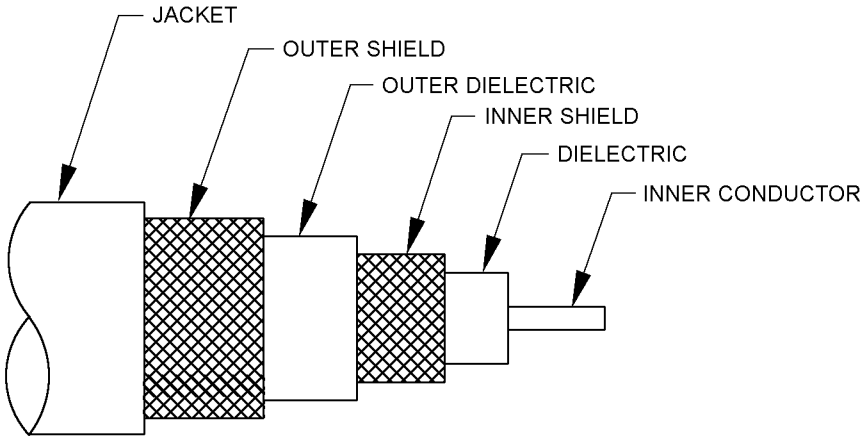


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Seven strands of AWG size 38 silver-coated, copper-covered steel. Overall diameter: .012 inch nominal.
Dielectric	Type F-1. Overall diameter: .033 ± .002 inch.
Inner shield	Silver-coated, AWG size 38 copper wire. Overall diameter: .048 - .054 inch. Coverage: 95.9% nominal Carriers: 16 Ends: 3 Picks/inch: 25.0 ± 10%
Outer dielectric	Type M-1. Overall diameter: .068 - .080 inch.
Outer shield	Silver-coated, AWG size 38 copper wire. Overall diameter: .084 - .098. Coverage: 95.9% nominal Carrier: 16 Ends: 6 Picks/inch: 13.4 ± 10%
Outer jacket	Type IX. Overall diameter: .104 - .128 inch. Wall thickness: .010 inch minimum.

AMSC N/A FSC 6145

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ENGINEERING INFORMATION:

Continuous working voltage:
750 V rms, minimum, inner conductor to inner shield.
300 V rms, minimum, inner shield to outer shield.
Operating frequency: 10 GHz, maximum.
Velocity of propagation: 69.5%, nominal.
Power rating: See figure 2.
Operating temperature range: -55°C to +200°C.
Inner conductor properties:
DC resistance (maximum at 20°C): 24.45 ohms per 100 feet.
Elongation: 10 percent, minimum.
Tensile strength: 50 $klb_f/inch^2$, minimum.

Engineering notes: This cable is useful in high temperature applications.

REQUIREMENTS:

Dimensions, configuration, and description: See figure 1 and table I.
Environmental and mechanical:
Visual and mechanical examination:
Out-of-roundness: Not applicable.
Eccentricity: Not applicable.
Adhesion of conductors:
Inner conductor to core: 1.5 pounds, minimum; 4 pounds, maximum.

Electrical and mechanical:
Operational:
Continuity: Applicable.
Spark test: 2,000 V rms +25 percent, -0 percent.
Voltage withstanding:
2,000 V rms, minimum, inner conductor to inner shield.
1,000 V rms, minimum, inner shield to outer shield.
Insulation resistance: Not applicable.
Corona extinction voltage: 1,000 V rms, minimum, inner conductor to inner shield.
Characteristic impedance: 50 ohms ± 2 ohms.

RF transmission loss (attenuation): 37 dB maximum at 400 MHz.
(This test is performed with the shield connected).
Standing wave ratio (return loss): See figure 3. (This test is performed with the shield connected).

Capacitance: 30.2 pF/ft, maximum, inner conductor to inner shield.
160 pF/ft, maximum, inner shield to outer shield.

Capacitance stability: Not applicable.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.
Aging stability: Not applicable.
Stress-crack resistance: +230° ± 5°C for 96 hours.
Outer conductor integrity: Not applicable.
Cold bend: -55° ± 2°C.

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Dimensional stability: +200° ± 5°C.
Inner conductor from core: .062 inch, maximum.
Inner conductor from jacket: .125 inch, maximum.
Contamination: Not applicable.
Bendability: Not applicable.
Flammability: Not applicable.
Flame propagation: Not applicable.
Acid gas generation: Not applicable.
Halogen: Not applicable.
Immersion tests: Not applicable.
Smoke index: Not applicable.
Toxicity index: Not applicable.

Durometer hardness: Not applicable.
Weathering: Not applicable.
Abrasion resistance: Not applicable.
Tear strength: Not applicable.
Heat distortion: Not applicable.
Physicals (aged): Not applicable.
Tensile strength and elongation: Not applicable.

Marking: See table II.
Weight: 1.65 pounds per 100 feet, maximum.
Workmanship: Not applicable.

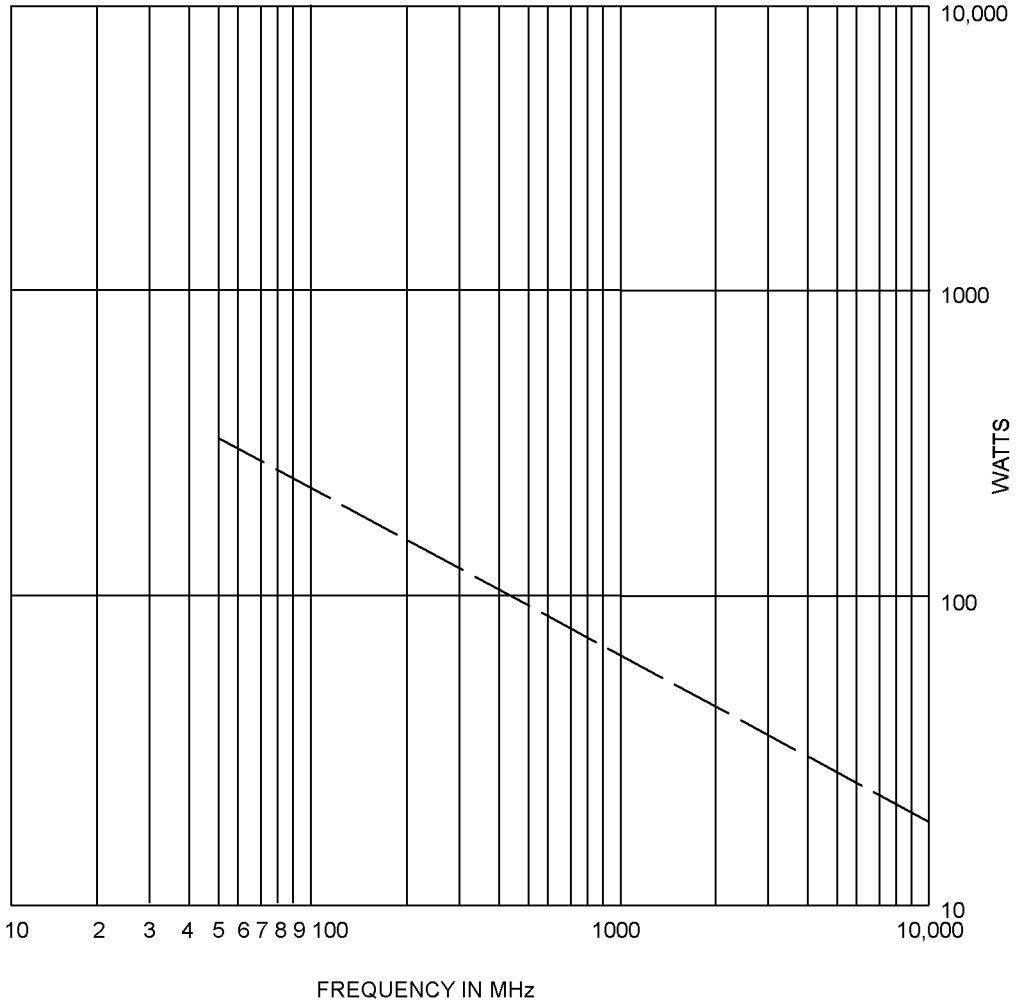
Part or Identifying Number (PIN): M17/131-RG403.

Supersession data: See table II.

TABLE II. Cross reference of PIN.

PIN	Superseded PIN or type designation
M17/131-RG403	RG-403/U

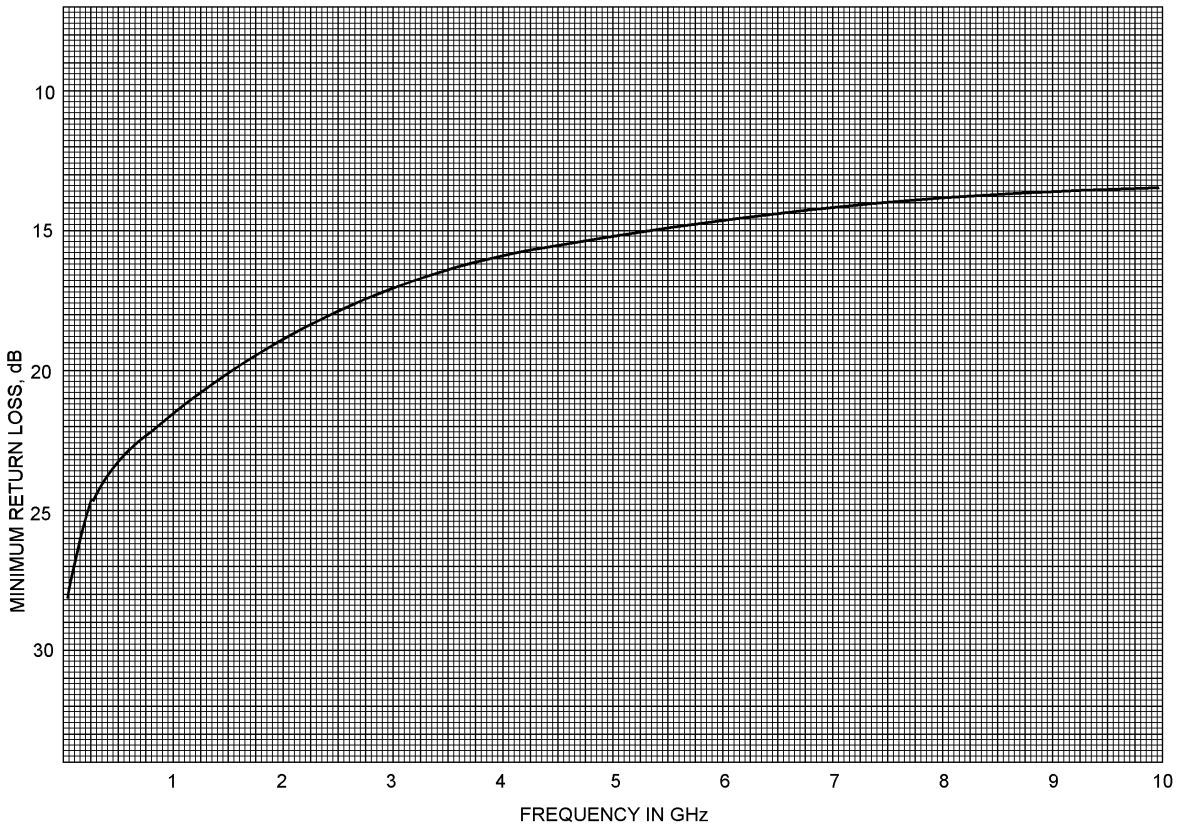
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POWER
WATTS
290
200
95
60
34
18

FIGURE 2. Power rating.

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SWR	REFLECTION COEFFICIENT	RETURN LOSS dB	SWR	REFLECTION COEFFICIENT	RETURN LOSS dB
17.3910	.8913	1	1.3767	.1585	16
8.7242	.7943	2	1.3290	.1413	17
5.8480	.7079	3	1.2880	.1259	18
4.4194	.6310	4	1.2528	.1122	19
3.5698	.5623	5	1.2222	.1000	20
3.0095	.5012	6	1.1957	.0891	21
2.6146	.4467	7	1.1726	.0794	22
2.3229	.3981	8	1.1524	.0708	23
2.0999	.3548	9	1.1347	.0631	24
1.9250	.3162	10	1.1192	.0562	25
1.7849	.2818	11	1.1055	.0501	26
1.6709	.2512	12	1.0935	.0447	27
1.5769	.2239	13	1.0829	.0398	28
1.4985	.1995	14	1.0736	.0355	29
1.4326	.1778	15	1.0653	.0316	30

Structural return loss	
MHz	dB
50	28
100	27
400	24
1000	22
3000	17
10000	14

FIGURE 3. Structural return loss.

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Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Referenced documents. This document references MIL-DTL-17.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2015-021)

Review activities:
Army – AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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